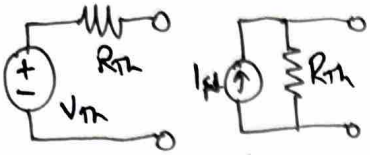
**Examination Aid Sheet**
**Faculty of Applied Science & Engineering**
Both sides of the sheet may be used;
must be printed on 8.5" x 11" paper.

Subject: ECE212 : CIRCUIT ANALYSIS
Candidate's name: ARNAV PATIL
Candidate's signature: Patil




$R_{Th} = \frac{V_{oc}}{I_{sc}}$ $P_{oc} = \frac{|V_T|^2}{8R_{Th}}$

$P_{max} = \frac{V_{oc} I_{sc}}{4} = \frac{V_{Th}^2}{4R_{Th}}$

Max Signal Transfer → $V_{max} = V_{oc}$, $I_{max} = I_{sc}$

Series RC → $R_T C v' + v = V_{Th}$ $\tau = RC = \frac{L}{R}$

Parallel RL → $\frac{L}{R_T} i' + i = i_{N0}$

$x(0^-) = x(0^+)$ → $x(t) = x(\infty) + (x(0^+) - x(\infty))e^{-t/\tau}$

Series RLC → $LCv_c'' + R_T C v_c' + v_c = V_{Th}$ | $\Delta = \zeta^2 - 1$

Parallel RLC → $LC i_L'' + \frac{L}{R} i_L' + i_L = i_{N0}$ | $\alpha = \zeta\omega_0, \beta = \omega_0\Delta$

$y'' + 2\zeta\omega_0 y' + \omega_0^2 y = 0$ → $\omega_0^2 = \frac{a_0}{a_2}$, $2\zeta\omega_0 = \frac{a_1}{a_2}$

Capacitors → $V_c(t) = \frac{1}{C}\int_0^t i(\tau)\,d\tau + V_c(0)$

↓

V lags I $i_c(t) = C\frac{dv_c}{dt}$

$V_R(s) = RI_R(s)$

$V_L(s) = LsI(s) - Li_L(0)$

$P_C = \frac{-\omega C V_s^2}{2}\sin(\omega t)$ $V_c(s) = \frac{1}{Cs}I_c(s) + \frac{V_c(0)}{s}$

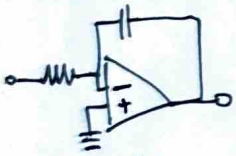
$U_C = \frac{CV^2}{2}$ From textbk, do not doubt

Inductors → $V_L(t) = L\frac{di}{dt}$ $i_L(t) = \frac{1}{L}\int_0^t v(\tau)\,d\tau + i_L(0)$

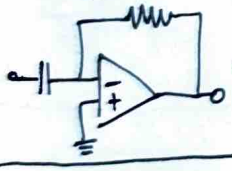
$P_L = \frac{-\omega L I_s^2}{2}\sin(\omega t)$ $U_L = \frac{LI^2}{2}$

OP AMPs → $A \to \infty$, $R_{in} \to \infty$, $R_{out} \to 0$, $V_o = A(V_p - V_n)$

Virtual short ONLY in negative feedback



$V_o = -\frac{1}{RC}\int_{t_0}^t v_i\,d\tau + V_o(t_0)$



$V_o = -RC\frac{dv_i}{dt}$

Integrating Factor

$y' + P(x)y = f(x)$

↳ $\mu(x) = e^{\int P(x)dx}$

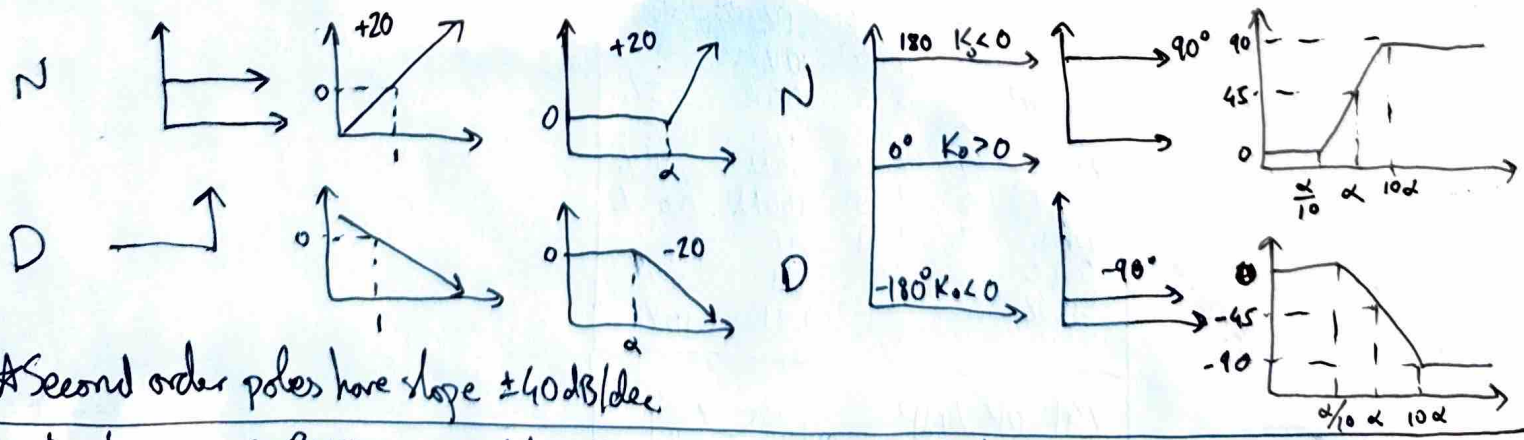
$\mu(x)y = \int \mu(x) f(x)\,dx$

Cramer's Rule

$$\begin{bmatrix} j2^A & 4^B \\ j6^C & j2^D \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 10\angle 0^\circ \; E \\ 0 \; F \end{bmatrix}$$

$I_1 = \frac{\Delta_1}{\Delta} = \frac{\begin{vmatrix} E & B \\ F & D \end{vmatrix}}{\begin{vmatrix} A & B \\ C & D \end{vmatrix}} = \frac{ED - BF}{AD - BC}$, $I_2 = \frac{AF - EC}{AD - BC}$

Gain $|K_0|_{dB}$ $|\omega|_{dB}$ $|(1 + j\omega/\alpha)|_{dB}$ Phase $\angle K_0$ $\pm 90^\circ$ $\tan^{-1}(\omega/\alpha)$




*Second order poles have slope ±40dB/dec

$\theta = \phi_v - \phi_i$ $S = P + jQ$ $Q = |S|\sin\theta$

$P = V_{rms} I_{rms}\cos\theta$ $|S| = V_{rms} I_{rms}$ $S = ZI_{rms}^2 = V_{rms}^2/Z^*$

$Q = V_{rms} I_{rms}\sin\theta$ $S = |S|\angle\theta$ $S = RI_{rms}^2 + jXI_{rms}^2$

$S = VI^*$ (rms) $P = |S|\cos\theta = |S| \times pf$

$Q_L = \omega L I_{rms}^2 = V_{rms}^2/\omega L$ Lag → $\theta > 0$ → inductive

$Q_C = -I_{rms}^2/\omega C = -\omega C V_{rms}^2$ Lead → $\theta < 0$ → capacitive

Max Power → $Z_L = Z_{Th}^*$

$T_V(s) = \frac{V_2(s)}{V_1(s)}$, $T_I(s) = \frac{I_2(s)}{I_1(s)}$ } Tranfer Function

$T_y(s) = \frac{I_2(s)}{V_1(s)}$ } Transadmittance

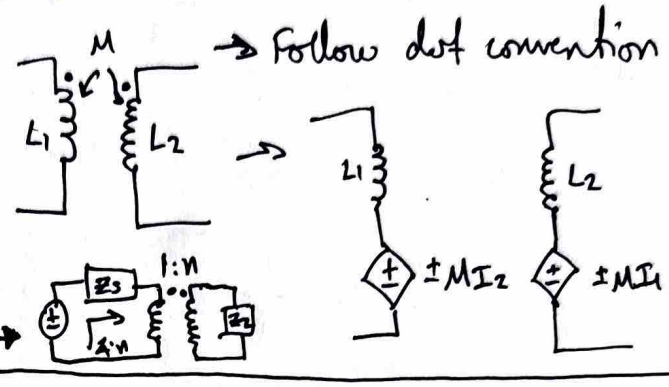
$T_z(s) = \frac{V_2(s)}{I_1(s)}$ } Transimpedance

$|T(j\omega_c)| = \frac{T_{max}}{\sqrt{2}}$, $|T(j\omega)|_{dB} = 20\log_{10}|T(j\omega)|$

$$\int_{0^-}^{0^+} (i'' + \cancel{i'} + \cancel{i})\,dt = \int_{0^-}^{0^+} \delta(t)\,dt$$

$i'(0^+) - i(0^-) = 1$

* When dealing w/ unit step or impulse functions




→ Follow dot convention

Reflected impedance → $Z_{in} = Z_L/n^2$

Low-Pass → $T(s) = \frac{K}{s+\alpha}$

High-Pass → $T(s) = \frac{Ks}{s+\alpha}$

$GBW = A\omega_c$

BANDPASS

$T(s) = T_1(s) \times T_2(s)$

↳ $\omega < \alpha_1 < \alpha_2$

$|T(j\omega)| = \frac{|K_1||K_2|\omega}{\alpha_1\alpha_2}$

↳ $\alpha_1 < \omega < \alpha_2$

$|T(j\omega)| = \frac{|K_1||K_2|}{\alpha_2}$

↳ $\alpha_1 < \alpha_2 < \omega$

$|T(j\omega)| = \frac{|K_1||K_2|}{\omega}$

BANDSTOP

$T(s) = T_1(s) + T_2(s)$

↳ $\omega < \alpha_1 < \alpha_2$

$|T(j\omega)| = \frac{|K_2|}{\alpha_2}$

$\boxed{Q = \frac{\omega_0}{B}}$

↳ $\alpha_1 < \omega < \alpha_2$

$|T(j\omega)| = 0$

↳ $\alpha_1 < \alpha_2 < \omega$

$|T(j\omega)| = |K_1|$

$\omega_c = \sqrt{\omega_{c1}\omega_{c2}}$

$B = \omega_{c2} - \omega_{c1} = R/L$

Series RLC Bandstop → $T(s) = \frac{V_2(s)}{V_1(s)} = \frac{Z_{LC}(s)}{R + Z_{LC}(s)}$

Parallel RLC Bandpass → $T(s) = \frac{I_2(s)}{I_1(s)} = \frac{1/R}{1/R + Y_{LC}(s)}$

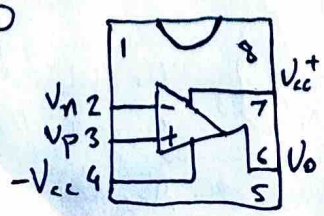
ACTIVE FILTERS → * high Q (steep roll-off) * no L * cascade design

LPF → $\frac{K}{(s/\omega_0)^2 + 2\zeta\omega_0 s + 1}$ HPF → $\frac{K(s/\omega_0)^2}{(s/\omega_0)^2 + 2\zeta\omega_0 s + 1}$

BPF → $\frac{K(s/\omega_0)}{(s/\omega_0)^2 + 2\zeta\omega_0 s + 1}$ BSF → $\frac{K[(s/\omega_0)^2 + 1]}{(s/\omega_0)^2 + 2\zeta\omega_0 s + 1}$

Capacitors → 10X ⇒ $10^x$ pF, Inductors → 10X ⇒ $10^x$ μH

| COLOR | 1 | 2 | 3 | MULT | TOL |
|---|---|---|---|---|---|
| BLACK | 0 | 0 | 0 | 1 | |
| BROWN | 1 | 1 | 1 | 10 | 1% |
| RED | 2 | 2 | 2 | 100 | 2% |
| ORANGE | 3 | 3 | 3 | 1K | |
| YELLOW | 4 | 4 | 4 | 10K | |
| GREEN | 5 | 5 | 5 | 100K | 0.5% |
| BLUE | 6 | 6 | 6 | 1M | 0.25% |
| VIOLET | 7 | 7 | 7 | 10M | 0.10% |
| GREY | 8 | 8 | 8 | 100M | 0.05% |
| WHITE | 9 | 9 | 9 | 1G | |
| GOLD | | | | 0.1 | 5% |
| SILVER | | | | 0.01 | 10% |

741 OP AMP

Pinout